

DATASET 3

CANCER STAGING AND DISEASE EXTENT — ONCOLOGY REFERENCE DATA

Abstract

Cancer staging is a standardized method used to describe the **anatomical extent of cancer at the time of diagnosis**. It provides a structured assessment of the primary tumor, involvement of regional lymph nodes, and presence or absence of distant metastasis. The **TNM classification system**, consisting of Tumor (T), Nodes (N), and Metastasis (M), is widely used for many solid tumors, with cancer-specific criteria determining how individual categories are defined.

Cancer staging supports treatment planning, prognostic assessment, communication between healthcare professionals, evaluation of treatment outcomes, clinical research, and population-level cancer-control activities. Clinical staging may be established using physical examination, imaging, endoscopy, biopsy, and other investigations, while pathological staging can provide additional information following surgical examination of tumor and lymph-node specimens.

Although anatomical stage is an important prognostic factor, it does not independently determine an individual's outcome. Tumor histopathology, grade, molecular and genetic characteristics, patient-related factors, treatment response, and other clinical variables may also influence prognosis and treatment selection. Therefore, cancer stage should be interpreted as one component of a broader oncological assessment.

1. Overview of Cancer Staging

Cancer staging is the process of determining the **anatomical extent of cancer** within the body at the time of diagnosis.

The assessment generally considers:

- The characteristics and local extent of the primary tumor.
- Involvement of regional lymph nodes.
- Presence or absence of distant metastasis.
- Extension into surrounding structures where relevant.

Cancer staging provides a common framework for describing disease extent and allows healthcare professionals to communicate information using standardized terminology.

Stage information can contribute to:

- Treatment planning.
- Prognostic assessment.
- Evaluation of treatment outcomes.
- Communication between treatment centers.
- Clinical and epidemiological research.
- Cancer-control planning.

Stage definitions need to remain sufficiently consistent to allow comparisons between populations over time, while also evolving when advances in medicine provide evidence that classifications should be modified.

2. General Cancer Stage Groups

Many solid tumors use stage groups represented by Roman numerals. However, the precise definitions differ between cancer types.

The following represents a **general framework and should not be treated as a universal staging rule for every malignancy**.

Stage 0 — In Situ Disease

Stage 0 generally describes abnormal or malignant cells that remain within the tissue layer in which they originated and have not invaded surrounding tissue.

General characteristics

- Abnormal cells remain localized.
- Invasive growth is absent.
- Distant metastasis is absent.
- May be described as **carcinoma in situ** where applicable.

Not all cancers have a Stage 0 classification.

Stage I — Localized / Early Disease

Stage I generally represents a relatively limited primary cancer that remains localized.

General characteristics

- Primary tumors are relatively limited.
- Disease remains primarily confined to the organ or tissue of origin.
- Distant metastasis is absent.
- Regional lymph-node involvement is generally absent in many Stage I classifications.

The exact definition depends on the cancer type.

Stage II — More Locally Advanced Disease

Stage II generally indicates a greater extent of primary disease than Stage I.

Possible characteristics include:

- Larger primary tumor.
- Greater depth of local invasion.
- Limited regional involvement in some cancer types.
- No distant metastasis within the relevant stage grouping.

The exact criteria vary considerably between cancers.

Stage III — Regional / Locally Advanced Disease

Stage III generally represents more extensive local or regional disease.

Possible characteristics include:

- Greater invasion into surrounding tissues.
- Regional lymph-node involvement.
- Extension into nearby structures.
- No distant metastasis in many Stage III classifications.

Some cancers subdivide Stage III into categories such as:

- IIIA
- IIIB
- IIIC

These subdivisions are cancer-specific.

Stage IV — Distant / Metastatic Disease

Stage IV generally indicates that cancer has spread to distant organs, tissues, or distant lymph nodes.

This is referred to as **metastatic cancer**.

Possible sites of distant spread include:

- Liver
- Lungs
- Bones
- Brain
- Other distant organs or tissues

For many solid tumors, confirmed distant metastasis corresponds to a Stage IV pattern.

3. TNM Classification System

The **TNM classification system** is widely used to describe the anatomical extent of many solid tumors.

The three principal components are:

T — Primary Tumor

N — Regional Lymph Nodes

M — Distant Metastasis

The individual categories are subsequently combined according to the cancer-specific staging system.

3.1 T — Primary Tumor

The **T category** describes the primary tumor.

Depending on the cancer type, it may consider:

- Tumor size.
- Depth of invasion.
- Local extension.
- Invasion of adjacent structures.
- Other cancer-specific characteristics.

Common notation

TX — Primary tumor cannot be adequately assessed.

To — No evidence of a primary tumor.

Tis — Carcinoma in situ where applicable.

T1–T4 — Increasing categories of tumor extent according to the specific cancer staging system.

The meaning of a particular T category differs between cancers.

3.2 N — Regional Lymph Nodes

The **N category** describes regional lymph-node involvement.

It can consider:

- Whether regional lymph nodes contain cancer.
- Number of affected lymph nodes.
- Location of involved nodes.
- Other cancer-specific characteristics.

Common notation

NX — Regional lymph nodes cannot be assessed.

No — No regional lymph-node involvement identified.

N₁, N₂, N₃ — Increasing categories of regional lymph-node involvement where applicable.

The criteria for N₁, N₂, and N₃ vary according to cancer type.

3.3 M — Distant Metastasis

The **M category** describes the distant spread of cancer.

Common notation

M₀ — No distant metastasis identified.

M₁ — Distant metastasis identified.

The M category is an important component of stage grouping in many solid tumors.

4. Important Principle of TNM Interpretation

TNM categories **cannot be interpreted identically across all cancers**.

For example, a T category may primarily describe tumor size in one cancer, while another cancer may define T according to depth of invasion or involvement of specific anatomical structures.

Similarly, the significance of regional lymph-node categories can differ according to:

- Number of involved nodes.
- Anatomical location.
- Size of nodal deposits.
- Cancer type.

Therefore:

TNM categories must always be interpreted according to the specific cancer and applicable staging classification.

5. Stage Grouping

After the T, N, and M categories have been established, they can be combined according to the appropriate cancer-specific staging system.

Simplified pathway

Primary Tumor » T Assessment



Regional Lymph Nodes » N Assessment



Distant Metastasis » M Assessment



T + N + M



Cancer-Specific Stage Group

Stage grouping may produce categories such as:

- Stage I
- Stage II
- Stage III
- Stage IV

Additional subdivisions may occur within a stage group, such as:

- IIA
- IIB
- IIIA
- IIIB
- IIIC

The precise combinations depend on the cancer type.

6. Clinical Staging — cTNM

Clinical staging is performed using information available before definitive treatment.

It may include:

- Physical examination.
- Imaging.
- Endoscopy.
- Biopsy.
- Laboratory investigations.
- Other diagnostic procedures.

Clinical staging is used to estimate the anatomical extent of disease before treatment and can help guide initial treatment planning.

Clinical notation

The prefix **c** may be used.

Examples:

cT₁

cN₀

cM₀

A complete clinical stage may therefore be represented by a combination such as:

cT₁N₀M₀

The exact stage group corresponding to this combination depends on the cancer type.

7. Pathological Staging — pTNM

Pathological staging is determined using information obtained from surgical specimens and histopathological examination when surgery provides sufficient material for assessment.

It may incorporate:

- Preoperative clinical findings.

- Surgical observations.
- Primary tumor examination.
- Histopathological examination.
- Regional lymph-node examination.

Pathological examination can identify microscopic disease that may not have been detectable through clinical examination or imaging.

Pathological notation

The prefix **p** may be used.

Examples:

pT2

pN1

A pathological classification may therefore be represented as:

pT2N1Mo

The precise stage group depends on the relevant cancer-specific staging system.

8. Clinical Stage Versus Pathological Stage

Clinical and pathological staging provide related but different information.

Feature	Clinical Staging	Pathological Staging
Timing	Before definitive treatment	Usually after surgical removal/specimen acquisition
Main evidence	Examination, imaging, biopsy and other tests	Surgical and histopathological findings
Prefix	c	p
Purpose	Initial assessment and treatment planning	More detailed assessment when surgical pathology is available
Example	cT2NoMo	pT2N1Mo

A pathological stage may differ from the original clinical assessment because examination of surgical tissue can reveal additional disease.

9. Post-Treatment Staging

Some patients receive treatment before surgery or definitive local treatment.

Such treatment can include:

- Chemotherapy.
- Radiation therapy.
- Targeted therapy.
- Other systemic treatment.

This approach may be used to reduce tumor burden or evaluate response before definitive treatment.

When staging is performed following certain preoperative treatments, a **y prefix** may be used.

Examples:

ycT1

ypT2

The prefix indicates that the classification was determined after a specified treatment course.

10. Recurrence and Restaging

Cancer may be reassessed if it:

- Recurs after treatment.
- Progress.
- Develops new sites of disease.
- Requires reassessment for subsequent treatment.

A lowercase **r** may be used for appropriate recurrence-related classifications.

Examples:

rcT1

rpT2

Restaging describes the disease at a later point in the clinical course.

Importantly, the original stage at diagnosis remains part of the patient's cancer history and is not simply replaced by a later disease status.

11. Routes of Cancer Spread

Cancer can extend beyond its original site through several mechanisms.

11.1 Direct Extension

The primary tumor may grow directly into surrounding tissues or anatomical structures.

This contributes to assessment of local tumor extent and may affect the T category.

11.2 Lymphatic Spread

Cancer cells can enter lymphatic vessels and reach regional lymph nodes.

Regional lymph-node involvement is evaluated within the **N category**.

The pattern of lymphatic spread differs between cancer types and anatomical sites.

11.3 Hematogenous Spread

Cancer cells can enter blood vessels and travel to distant organs.

When cancer establishes disease at a distant anatomical location, this is known as **metastasis**.

Distant metastatic disease is represented by the **M category**.

12. Investigations Used in Cancer Staging

Cancer staging generally requires information from multiple investigations rather than a single test.

12.1 Physical Examination

Physical examination can provide information about:

- Tumor location.
- Palpable masses.
- Enlarged lymph nodes.
- Local anatomical changes.

Its usefulness depends on the cancer type and anatomical location.

12.2 Computed Tomography — CT

CT imaging can provide detailed anatomical information about:

- Primary tumors.
- Regional lymph nodes.
- Surrounding structures.
- Possible distant disease.

CT is commonly incorporated into staging investigations for many cancers.

12.3 Magnetic Resonance Imaging — MRI

MRI provides detailed soft-tissue imaging and may be particularly useful for certain anatomical regions and tumor types.

It can assist in evaluating:

- Local tumor extension.
- Invasion of surrounding structures.

- Selected metastatic sites.
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12.4 Ultrasound

Ultrasound can be used to evaluate selected organs, masses, and regional structures.

It may also assist in guiding some tissue-sampling procedures.

12.5 PET / PET-CT

PET imaging can identify areas of increased metabolic activity and may be used in selected cancers to evaluate:

- Primary disease.
 - Regional disease.
 - Possible distant metastasis.
 - Treatment response in appropriate circumstances.
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12.6 Endoscopy

Endoscopy allows direct examination of the interior of certain organs.

It may provide information about:

- Tumor location.
- Local extent.
- Visible abnormalities.

Tissue samples can often be obtained during the procedure.

12.7 Biopsy

A biopsy involves obtaining tissue for microscopic examination.

Biopsy can help:

- Confirm malignancy.
- Determine histological type.
- Assess tumor characteristics.
- Evaluate suspicious lesions.

Biopsy alone does not necessarily provide the complete anatomical stage.

12.8 Blood and Laboratory Tests

Blood tests may provide supportive information in selected cancers.

Certain **tumor markers** can be useful for:

- Supporting diagnosis in appropriate contexts.
- Assessing disease burden in selected cancers.
- Monitoring treatment response.
- Detecting possible recurrence.

Tumor markers generally should not be interpreted in isolation as a complete staging method.

13. SEER Summary Stage

A simplified system called **SEER Summary Stage** is used for population-level cancer statistics and surveillance.

Its broad categories include:

In Situ

Abnormal or malignant cells remain within the tissue layer in which they originated and have not invaded surrounding tissue.

Localized

Cancer remains within the organ where it originated.

Regional

Cancer has extended beyond the original site into nearby tissues, structures, or regional lymph nodes without identified distant spread.

Distant

Cancer has spread to distant organs, structures, or distant lymph nodes.

Unknown

Insufficient information is available to determine the extent of disease.

This type of simplified classification can support population-level comparison and cancer surveillance.

14. Other Cancer Staging Systems

TNM is not appropriate for every malignancy.

Some cancers use other classification systems because their biology and patterns of spread differ from those of conventional solid tumors.

Examples include:

Lugano Classification

Used for staging certain lymphomas, including Hodgkin lymphoma and non-Hodgkin lymphoma.

FIGO Staging

Used for cancers of the female reproductive system.

Hematological Malignancies

Many blood cancers, such as leukemia, are not staged using conventional solid-tumor TNM classification.

Brain and Central Nervous System Tumors

Many tumors of the brain and central nervous system do not use conventional TNM staging because they generally do not behave like tumors that spread through regional lymph nodes in the same way as many other solid cancers.

15. Tumor Profile and Anatomical Stage

An important distinction exists between **anatomical stage** and **tumor profile**.

Anatomical Stage

Describes:

Where the cancer is and how far it has spread.

The main framework is:

T + N + M

Tumor Profile

Describes characteristics of the cancer itself.

These may include:

- Histological type.
- Morphology.
- Tumor grade.
- Molecular characteristics.
- Genetic characteristics.
- Biomarker status.

These categories provide different types of information.

Anatomical stage should therefore not automatically be combined with every molecular or genetic characteristic into a single definition of "stage."

16. Tumor Grade

Tumor **grade** describes how abnormal cancer cells appear under microscopic examination and may provide information about biological behavior.

A general classification may include:

Low Grade

Cells may retain more characteristics resembling the normal tissue from which they originated.

Intermediate Grade

Cells show a greater degree of abnormality.

High Grade

Cells may show substantial abnormalities and loss of characteristics of the original tissue.

Grade and stage are **not the same measurement**.

Stage » Anatomical extent

Grade » Microscopic appearance / differentiation

Both may contribute to prognostic assessment.

17. Factors Influencing Prognosis

Cancer stage is an important prognostic factor, but it does not independently determine an individual's outcome.

Relevant factors may include:

Disease-Related Factors

- Anatomical extent.
- Tumor size.
- Local invasion.
- Lymph-node involvement.

- Distant metastasis.
- Histological type.
- Tumor grade.
- Molecular characteristics.
- Genetic characteristics.

Patient-Related Factors

- Age.
- General health.
- Functional status.
- Other medical conditions.
- Relevant immune status.

Treatment-Related Factors

- Type of treatment.
- Timing of treatment.
- Response to treatment.
- Ability to complete treatment.

Healthcare-Related Factors

- Access to diagnostic services.
- Availability of treatment.
- Availability of specialist expertise.
- Supportive and palliative-care resources.

These factors can contribute to differences in outcomes among patients with the same anatomical stage.

18. Cancer Stage and Prognosis

In general, cancers diagnosed at a more localized stage often have more favorable outcomes than cancers diagnosed after regional or distant spread.

However, survival statistics should be interpreted carefully.

A survival statistic represents the outcomes observed in a population of people with a particular cancer or disease characteristic. It does **not** provide a guaranteed prediction for an individual patient.

Therefore:

Cancer Stage $\neq$ Individual Prognosis

A more appropriate conceptual model is:

Stage + Tumor Biology + Patient Factors + Treatment + Treatment Response $\gg$ Prognostic Assessment

The same stage can therefore be associated with different outcomes in different patients.

19. Stage and Treatment Planning

Cancer stage can influence the treatment strategy selected by the clinical team.

Depending on cancer type and stage, treatment may involve one or more of:

- Surgery.
- Radiation therapy.
- Chemotherapy.
- Immunotherapy.
- Targeted therapy.
- Hormone therapy.
- Other systemic therapies.
- Supportive or palliative care.

Treatment selection also depends on factors beyond anatomical stage, including tumor biology, molecular characteristics, patient health, treatment availability, and patient-specific considerations.

20. Early-Stage and Advanced-Stage Disease

Early / Localized Disease

Localized disease generally remains confined to the site where the cancer originated.

Potential characteristics:

- Limited tumor extent.
 - No distant metastasis.
 - Greater emphasis on local treatment in cancers where local therapy is appropriate.
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Regional Disease

Regional disease has extended beyond the primary site into nearby structures and/or regional lymph nodes.

Potential characteristics:

- Local invasion.
 - Regional lymph-node involvement.
 - Increased anatomical extent compared with localized disease.
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Distant / Metastatic Disease

Distant disease involves spread beyond the primary and regional sites.

Potential characteristics:

- Distant organ involvement.
- Distant lymph-node involvement where applicable.
- M1 classification in many TNM systems.

Treatment may focus on controlling disease throughout the body, treating specific metastatic sites when appropriate, relieving symptoms, and maintaining quality of life.

21. Brief Comparative Oncology Case Studies

The following are **simplified educational examples** designed to demonstrate staging concepts. They do not represent complete clinical staging protocols.

Case Study 1 — Localized Breast Tumor

A patient is found to have a relatively small breast tumor. Imaging does not identify regional lymph-node involvement or distant metastasis.

Observed findings

Primary tumor: Present

Regional nodes: No identified involvement

Distant metastasis: Not identified

General interpretation

The findings demonstrate a localized disease pattern.

A formal stage would require breast-cancer-specific TNM criteria and relevant pathological and biological information.

Case Study 2 — Regional Lung Cancer

A patient has a lung tumor with involvement of regional lymph nodes. Investigations do not identify distant metastatic disease.

Observed findings

Primary tumor: Present

Regional nodes: Involved

Distant metastasis: Not identified

General interpretation

The disease demonstrates regional progression.

The exact stage requires the appropriate lung-cancer TNM classification.

Case Study 3 — Locally Advanced Colorectal Cancer

A colorectal tumor extends through the bowel wall, and regional lymph nodes contain cancer cells. No distant metastasis is identified.

Observed findings

Local invasion: Present

Regional lymph nodes: Involved

Distant metastasis: Mo

General interpretation

The findings represent regional or locally advanced disease.

The precise stage requires the pathological T and N categories and colorectal-cancer-specific stage grouping.

Case Study 4 — Metastatic Colorectal Cancer

A patient is diagnosed with a colorectal primary tumor. Subsequent investigation confirms cancer involvement in the liver.

Observed findings

Primary tumor: Colorectal

Distant organ: Liver

Distant metastasis: M1

General interpretation

Confirmed distant metastasis represents metastatic disease and generally corresponds to a Stage IV pattern in colorectal cancer.

22. Comparative Disease-Extent Framework

Disease Extent	Primary Tumor	Regional Nodes	Distant Spread
In Situ	Abnormal cells confined to tissue layer	Not applicable/absent	Absent
Localized	Confined to primary site	Usually absent	Absent
Regional	Present with local/regional extension	May be involved	Not identified
Distant	Present	May or may not be involved	Present
Unknown	Insufficient information	Insufficient information	Cannot be adequately determined

This framework is useful for describing broad disease extent but does not replace cancer-specific TNM staging.

23. Why Cancer Stage Is Important in Research

Consistent cancer staging allows researchers to compare groups of patients with similar disease extent.

Stage information can support:

- Clinical trials.
- Treatment-outcome studies.
- Survival analysis.
- Epidemiological research.
- Cancer registry analysis.
- Population-level cancer surveillance.
- Evaluation of cancer-control programs.

Recording stage alongside cancer incidence provides more information about the burden and characteristics of cancer within a population.

24. Cancer Stage and Cancer Control

Stage information can help identify whether a population is predominantly presenting with localized, regional, or advanced disease.

A population with a high proportion of early-stage diagnoses may require greater emphasis on:

- Diagnostic services.
- Screening.
- Surgical services.
- Radiation facilities.

A population with a high proportion of metastatic disease may require greater investment in:

- Systemic cancer treatment.
- Palliative care.
- Supportive care.
- Symptom management.

Therefore, stage data can contribute to health-system planning and cancer-control strategies.

25. Important Limitations of Cancer Staging

Cancer staging provides valuable information, but several limitations should be recognized.

Cancer-Specific Definitions

Stage categories differ between cancer types.

Incomplete Information

Sometimes the available clinical information is insufficient to establish the complete extent of disease.

Imaging Limitations

Small metastatic deposits or microscopic disease may not always be detectable using imaging.

Biological Variation

Two tumors with similar anatomical stages can have different biological behaviors.

Patient Variation

Patients with the same cancer stage can have different overall health, treatment tolerance, and treatment responses.

Changing Classification Systems

Staging systems can be revised as new evidence and medical technologies become available.

Therefore, stages should always be interpreted in the context of the appropriate cancer-specific classification system and clinical information.

26. Key Oncology Data Points

Cancer Stage

Describes the **anatomical extent of disease**.

T

Describes the **primary tumor**.

N

Describes **regional lymph-node involvement**.

M

Describes **distant metastasis**.

cTNM

Clinical TNM determined before definitive treatment.

pTNM

Pathological TNM determined using surgical and histopathological findings.

yTNM

TNM classification following certain preoperative treatments.

rTNM

TNM classification relating to recurrent disease where applicable.

Stage Group

A cancer-specific grouping derived from TNM and, where applicable, other required criteria.

Grade

Describes microscopic characteristics and differentiation of tumor cells.

Prognosis

Expected outcome influenced by stage and multiple additional disease, patient, and treatment factors.

27. Key Findings

The combined oncology literature supports the following conclusions:

1. Cancer staging describes the **anatomical extent of disease**.
2. The TNM system is widely used for many solid tumors.
3. **T** represents the primary tumor.
4. **N** represents regional lymph-node involvement.
5. **M** represents distant metastasis.
6. TNM categories are **cancer-specific** and cannot be interpreted identically across all malignancies.
7. Clinical staging uses information available before definitive treatment.

8. Pathological staging can provide additional information following surgery and histopathological examination.
 9. Cancer may be reassessed following treatment or recurrence.
 10. The original stage at diagnosis remains important for interpreting disease history and population statistics.
 11. Cancer can spread through direct extension, lymphatic pathways, and blood vessels.
 12. CT, MRI, PET, ultrasound, endoscopy, biopsy, physical examination, and selected laboratory tests may contribute to staging.
 13. Not all cancers use conventional TNM staging.
 14. Stage 0–IV groupings provide a general framework, but exact definitions vary by cancer type.
 15. Stage is an important prognostic factor but is **not an independent prediction of an individual's outcome**.
 16. Tumor grade, histological type, molecular characteristics, genetic features, patient characteristics, and treatment response can also influence prognosis.
 17. Stage data are valuable for treatment planning, clinical research, cancer registries, and population-level cancer-control activities.
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Conclusion

Cancer staging is a fundamental component of oncology because it provides a standardized description of the anatomical extent of malignant disease. The TNM system organizes information regarding the primary tumor, regional lymph nodes, and distant metastasis, while cancer-specific stage groups provide a broader classification of disease extent.

Clinical staging and pathological staging provide different levels of information, and additional assessments may be required after treatment or recurrence. Imaging, pathology, physical examination, endoscopy, and laboratory investigations can all contribute to determining disease extent.

Although stage is an important indicator of prognosis and treatment strategy, it should not be interpreted as an isolated prediction of individual outcome. Tumor biology, histopathology, grade, molecular and genetic characteristics, patient factors, treatment, and response to treatment can all influence the clinical course.

Cancer staging therefore represents one important component of comprehensive oncological assessment, providing a structured framework for understanding disease extent, planning treatment, evaluating outcomes, conducting research, and monitoring cancer at the population level.

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